3 **(a)** State the principle of conservation of momentum.

<u>sum / total</u> momentum before (a collision) = <u>sum / total</u> momentum after (a collision) **or** <u>sum / total</u> momentum (of a system) is constant	**M1**
if no (resultant) external force (acts) / for an isolated system	**A1**

.... [2]

(b) An object of mass $2m$ is travelling at a speed of $5.0\,m\,s^{-1}$ in a straight line. It collides with an object of mass $3m$ which is initially stationary, as shown in Fig. 3.1.

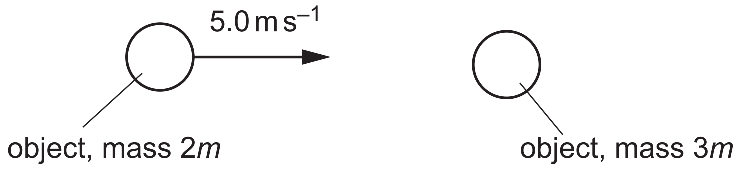


Fig. 3.1

After the collision, the object of mass $2m$ moves with velocity v at an angle of 30° to its original direction of motion.

The object of mass $3m$ moves with velocity w also at an angle of 30°, as shown in Fig. 3.2.

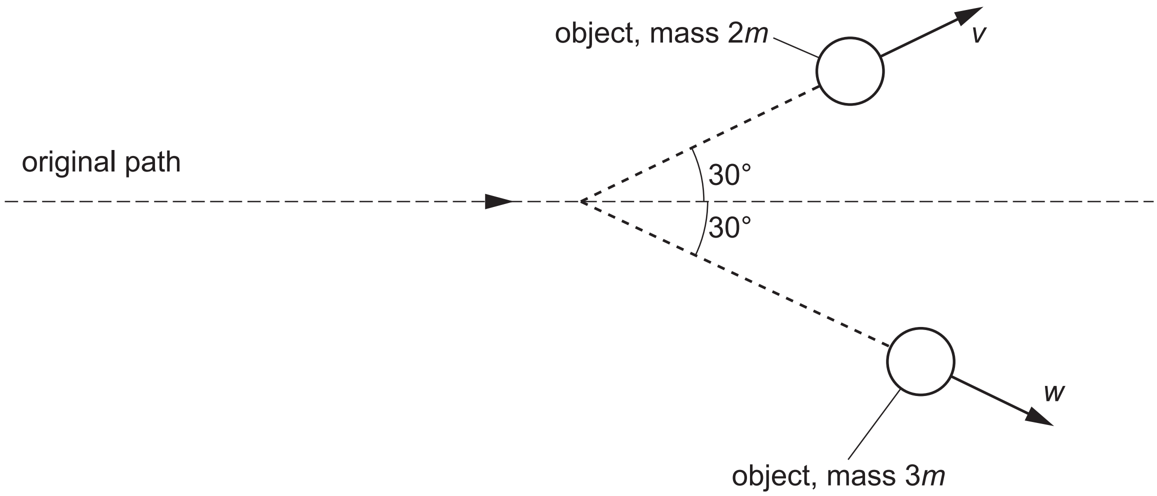


Fig. 3.2

By considering the conservation of momentum in two dimensions, calculate the magnitudes of v and w.

along direction of motion: $10m = 2mv \cos 30° + 3mw \cos 30°$	**C1**
perpendicular to direction of motion: $2mv \cos 60° = 3mw \cos 60°$ ($v = 3w / 2$)	**C1**
$v = 2.9\ \text{m s}^{-1}$	**A1**
$w = 1.9\ \text{m s}^{-1}$	**A1**

v = ... m s^{-1}

w = ... m s^{-1}
[4]

(c) An object of mass 4.2 kg is travelling in a straight line at a speed of $6.0\ \text{m s}^{-1}$. The object is brought to rest in a distance of 0.050 m by a constant force.

Calculate the magnitude of this force.

$E_K = ½ \times m \times v^2$ $(= ½ \times 4.2 \times 6.0^2)$ (= 76 J)	**C1**
force = work done / distance	**C1**
force = 76 / 0.050 = 1500 N	**A1**

or	
$a = (–)u^2 / 2s$ $= (–)6.0^2 / (2 \times 0.050)$ $(= (–)360\ \text{m s}^{-2})$	**(C1)**
$F = ma$	**(C1)**
$F = 4.2 \times 360$ = 1500 N	**(A1)**
or	
$a = (–)u^2 / 2s$ $= (–)6.0^2 / (2 \times 0.050)$ $(= (–)360\ \text{m s}^{-2})$	**(C1)**
$t = –u / a$ $= –6.0 / –360$ (= 0.017 s) $F = \Delta p / t$	**(C1)**
$F = (0 – 4.2 \times 6) / 0.017$ = 1500 N	**(A1)**

... N [3]

[Total: 9]

2 **(a)** Define linear momentum.

product of mass and velocity	**B1**

.... [1]

(b) A car of mass 1800 kg is moving in a straight line. Fig. 2.1 shows the variation with time *t* of the momentum *p* of the car.

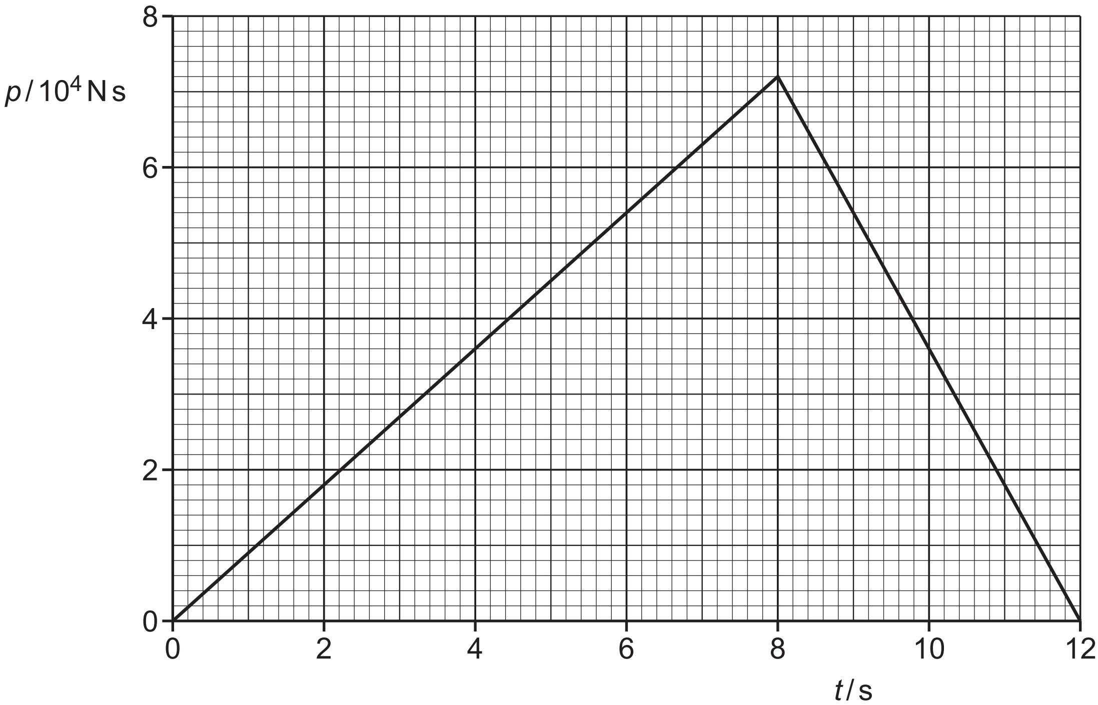


Fig. 2.1

(i) Calculate the maximum speed reached by the car.

maximum speed $= 7.2 \times 10^4 / 1800$ $= 40\ \text{m s}^{-1}$	**A1**

maximum speed = ... m s^{-1} [1]

(ii) Calculate the maximum kinetic energy of the car.

kinetic energy $= \frac{1}{2} mv^2$	**C1**
$= \frac{1}{2} \times 1800 \times 40^2$ $= 1.4 \times 10^6$ J	**A1**

maximum kinetic energy = ... J [2]

(iii) Show that the acceleration of the car at time $t = 4.0\,\text{s}$ is $5.0\,\text{m}\,\text{s}^{-2}$.

a = gradient of line / mass **or** $a = (v - u) / t$ **or** $a = \Delta v / t$	**C1**
a = e.g. $(3.6 \times 10^4) / (4.0 \times 1800) = 5.0\,\text{m}\,\text{s}^{-2}$ **or** a = e.g. $20\ (-\ 0) / 4 = 5.0\,\text{m}\,\text{s}^{-2}$ **or** F = e.g. $3.6 \times 10^4 / 4.0 = 9.0 \times 10^3$ **and** $a = 9.0 \times 10^3 / 1800 = 5.0\,\text{m}\,\text{s}^{-2}$	**A1**

[2]

(iv) Determine the distance travelled by the car between times $t = 0$ and $t = 12.0\,\text{s}$.

distance = $\frac{1}{2} \times 40 \times (8 + 4)$ **or** distance = $\frac{1}{2} \times 72\,000 \times (8 + 4) / 1800$ **or** distance = $[40^2 / (2 \times 5)] + [40^2 / (2 \times 10)]$ **or** distance = $\frac{1}{2} \times 5 \times 8^2 + (40 \times 4 - \frac{1}{2} \times 10 \times 4^2)$	**C1**
distance = 240 m	**A1**

distance = ... m [2]

(c) On Fig. 2.2, sketch the variation with time t of the acceleration a of the car in **(b)** from $t = 0$ to $t = 12.0\,\text{s}$.

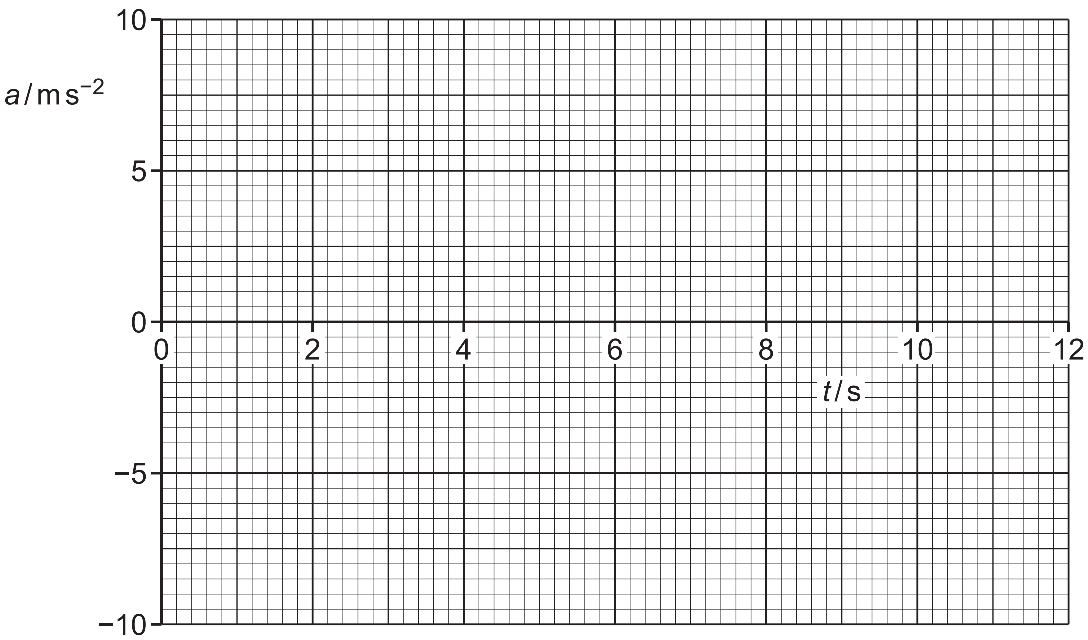


Fig. 2.2

[3]

[Total: 11]

stepped shape, showing one constant value up to 8.0 s then stepping to a different constant value from 8.0 s with no time delay	**B1**
horizontal straight line from (0, 5.0) to (8.0, 5.0)	**B1**
horizontal straight line from (8.0, –10.0) to (12.0, –10.0)	**B1**

2 **(a)** Define momentum.

product of mass and velocity	B1

.... [1]

(b) A child stands on a scooter on horizontal ground. The combined mass of the child and the scooter is 16 kg.
The child starts from rest and pushes once on the ground with her foot which causes her to accelerate. The push lasts for a time of 1.1 s. The speed of the child and the scooter after the push is $0.60\,m\,s^{-1}$.

Determine the average resultant force acting horizontally on the child and the scooter during the push.

$F = m\Delta v / \Delta t$ $= 16 \times 0.60 / 1.1$	C1
$= 8.7$ N	A1

average force = ... N [2]

(c) Later, the child in **(b)** travels down a slope at a constant angle to the horizontal, as shown in Fig. 2.1.

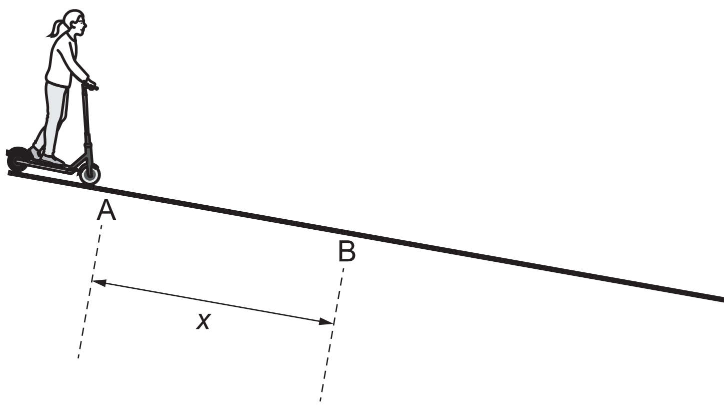


Fig. 2.1 (not to scale)

At point A her speed is $0.60\,m\,s^{-1}$. She has a constant acceleration of $0.85\,m\,s^{-2}$ parallel to the slope. After a time of 3.7 s, she reaches point B.

Calculate the distance x travelled by the child along the slope from A to B.

$x = ut + ½at^2$ $x = 0.60 \times 3.7 + ½ \times 0.85 \times 3.7^2$	C1
or	
$v = 0.60 + 0.85 \times 3.7$ (= $3.75\,m\,s^{-1}$) $x = 3.75 \times 3.7 - 0.5 \times 0.85 \times 3.7^2$ **or** $x = ½ \times (0.60 + 3.75) \times 3.7$ **or** $x = (3.75^2 - 0.60^2) / (2 \times 0.85)$	(C1)
$x = 8.0$ m	A1

m [2]

(d) At point B, the child in **(c)** applies the brake with a constant force to maintain a constant velocity. Point C is 18 m from point B, as shown in Fig. 2.2.

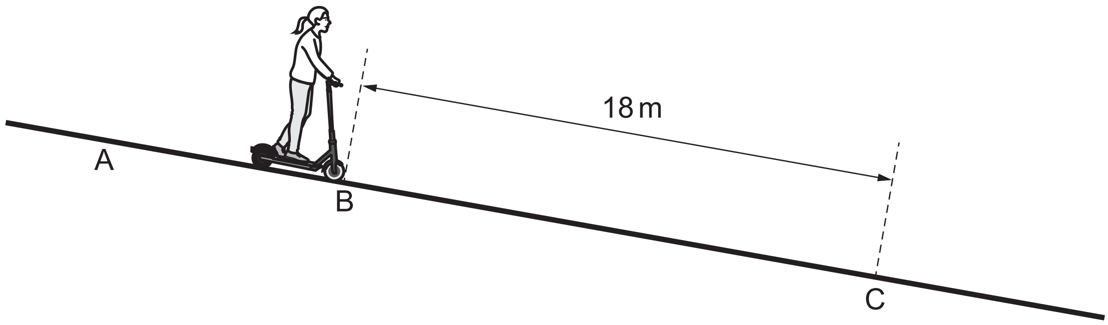


Fig. 2.2 (not to scale)

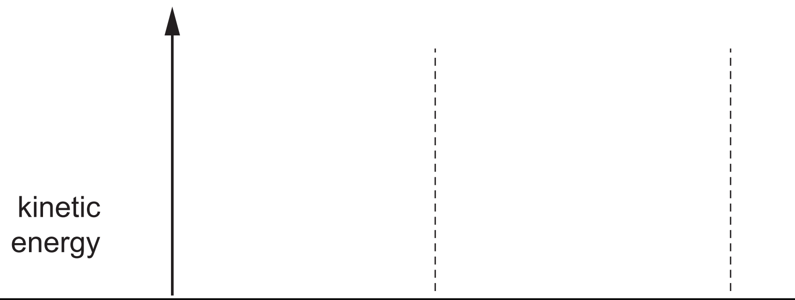
The work done by the braking force between B and C is 250 J.

(i) Determine the magnitude of the braking force.

$F = W / s$	**C1**
$= 250 / 18$ $= 14$ N	**A1**

force = ... N [2]

(ii) On Fig. 2.3, sketch the variation of the kinetic energy of the child and scooter with distance travelled from point A to point C.
Numerical values for kinetic energy are not required.



any line starting at distance = 0 and a positive non-zero value of kinetic energy	**B1**
a straight line from distance = 0 to distance = x with positive gradient	**B1**
a straight horizontal line at a non-zero value of kinetic energy starting at distance = x and ending at distance = x + 18 m that is continuous with the previous line	**B1**

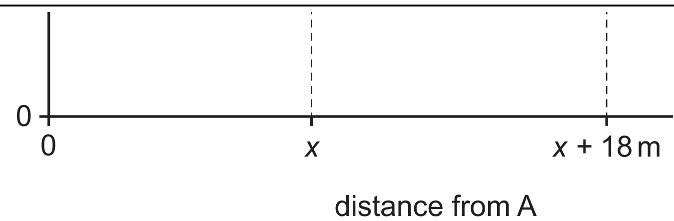


Fig. 2.3

[3]

[Total: 10]

2 **(a)** State the principle of conservation of momentum.

sum / total momentum (of a system of bodies) is constant **or** sum / total momentum before = sum / total momentum after	**M1**
for an isolated system / no (resultant) external force	**A1**

[2]

(b) A ball X has mass 240 g and moves in a straight line on a horizontal frictionless surface with an initial speed of $16\,\text{m}\,\text{s}^{-1}$. The ball collides with a stationary ball Y that has mass 480 g. After the collision, ball X is stationary, as shown in Fig. 2.1.

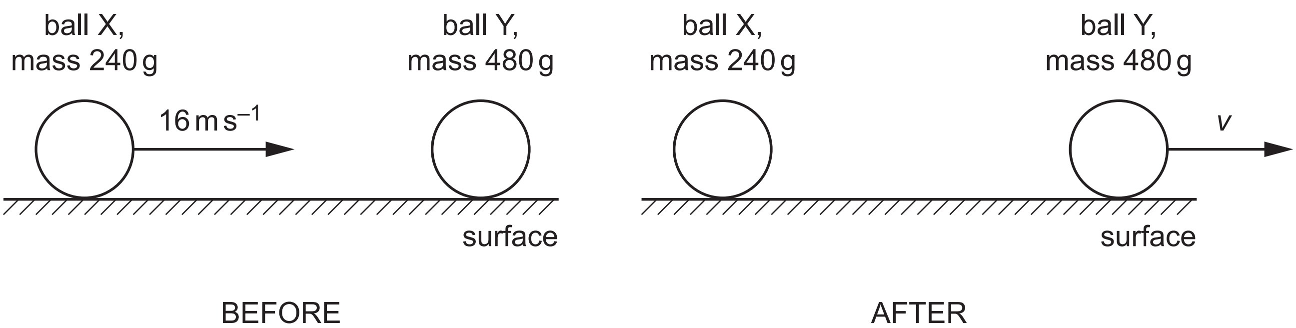


Fig. 2.1

(i) Show that the speed v of ball Y after the collision is $8.0\,\text{m}\,\text{s}^{-1}$.

$240 \times 16 = 480v$ and so $v = 8.0\,\text{m}\,\text{s}^{-1}$ **or** (initial momentum =) 240×16 (= $3840\,\text{g}\,\text{m}\,\text{s}^{-1}$) **and** $v = 3840 / 480 = 8.0\,\text{m}\,\text{s}^{-1}$	**A1**

[1]

(ii) Calculate the change in the total kinetic energy ΔE_K of the balls due to the collision.

(E_K =) $\frac{1}{2}mv^2$	**C1**
$\Delta E_K = \frac{1}{2}[(0.24 \times 16^2) - (0.48 \times 8.0^2)]$	**C1**
= 15 J	**A1**

ΔE_K = ... J [3]

(c) The collision in **(b)** lasts for a time of 2.0 ms. Assume that the contact force between the balls is constant during this time.

(i) Determine the magnitude and direction of the force exerted on ball X by ball Y during the collision.

$F = (0.24 \times 16) / (2.0 \times 10^{-3})$ **or** $F = (0.48 \times 8) / (2.0 \times 10^{-3})$	**C1**
= 1900 N	**A1**
direction: to the left	**B1**

magnitude = ... N

direction ....

[3]

(ii) Compare the magnitude and direction of the force exerted on ball Y by ball X during the collision with the answers in **(c)(i)**. No further calculations are required.

equal (magnitude)	**B1**
opposite (direction)	**B1**

.... [2]

[Total: 11]